

Volume 2 — Operation Manual (LinuxCNC)

Mazak VQC-20/40 SN 060231 — LinuxCNC/Mesa Retrofit

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Applicability: This volume describes operation **after** bring-up is complete and all Volume 3 verification checklists are signed off. During bring-up, follow Volume 3 procedures instead.

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Chapter 1 — Control Station Overview

The Mazatrol CRT and keyboard are replaced by a **LinuxCNC PC**. The machine configuration is `mazak_vqc_20_40.ini`, which loads the HAL files that map the Mesa hardware:

- `motion_7i97t.hal` — axes, spindle command, limits/homes/E-stop
- `field_7i84u.hal` — ATC, magazine, coolant, lube, hydraulics, panel I/O
- `postgui.hal` / `pendant_whb04b.hal` — GUI extras and optional pendant

Physical operator devices that may be retained on the original panel (wired to the 7i84U): **Cycle Start** (IN28), **Feed Hold** (IN29), **Single Block** (IN30), plus the alarm light/horn (OUT15). All are marked optional-reuse in the pin authority table.

Chapter 2 — Power-Up Sequence

1. Close the cabinet; verify no lockout tags.
2. Turn on the main disconnect (460 VAC). The transformer energizes the 230 VAC bus and the HR-11F-24 24 VDC supply.
3. Wait for the hydraulic pump if it is on a delay; confirm hydraulic pressure OK (`hydraulic-ok` , Sanwa SPS-8T-PC-20).
4. Power on the LinuxCNC PC. The 7i97T (192.168.1.121) is discovered over Ethernet at LinuxCNC start.
5. Launch LinuxCNC and select the `mazak_vqc_20_40` configuration.
6. Release/reset all E-stops, then press **F1 (E-stop reset)** and **F2 (Machine On)**.
7. Confirm the drive enables pick (S-ON to each TRA drive) and no drive faults show.

If Machine On fails: check the E-stop chain state (Ch. 12), drive fault inputs, and hydraulic pressure — all are preconditions.

Chapter 3 — Homing & Referencing

Axis position comes from the Tamagawa resolvers through the Mesa 7i49. Resolvers are absolute **within one revolution only**, so each axis must reference against its home switch after power-up.

1. Clear the table; ensure the Z axis has room to rise.
2. Home **Z first**, then X, then Y (keeps the head clear of fixtures).
3. Use the homing command in the GUI (Home All if configured; otherwise per-axis).
4. Watch that each axis moves *toward* its home switch (TB5.7–TB5.9). If any axis moves the wrong way, stop — the homing setup needs review (Vol 5 §6).
5. Verify machine zero against a known reference before trusting offsets.

Chapter 4 — Jogging & Manual Control

- Select jog increments (continuous / 1.0 / 0.1 / 0.01 / 0.001) before touching a jog key.
- Keyboard arrows / Page keys jog X, Y, Z in the standard LinuxCNC layout; the pendant (Ch. 10) may also be used.
- Overtravel limits (TB5.1–TB5.6) will halt motion, but they are the last resort — jog with your eyes on the machine, not the screen.
- Rapid-speed jogs only with the work area clear.

Chapter 5 — Spindle Operation

The Mitsubishi FR-SX drive receives its speed reference from the 7i97T analog output (TB3.15/16) and its run/direction/enable commands from the 7i84U (OUT0 FWD, OUT1 REV, OUT2 ENA).

- **M3 Sxxxx** — spindle forward at S RPM; **M4** — reverse; **M5** — stop.
- **Gear ranges:** the hydraulic Hi/Lo gearbox is commanded by `gear-hi-sol` / `gear-lo-sol` with confirmation switches PRS-10 / PRS-12. Allow gear shifts to complete (confirm input true) before loading the spindle.
- **At-speed:** feed moves that require the spindle wait on `spindle-at-speed` (FR-SX speed-reach output, IN13).
- **Orient:** spindle orient is required for tool change; do not command an ATC cycle unless orient completes.
- Do not command speed changes across a gear range mid-cut.

Chapter 6 — Tool Changes & the ATC / Magazine

The hydraulic ATC and BCD-coded magazine are retained from the original machine.

Automatic (M6 Tn): LinuxCNC sequences magazine position (BCD bits PRS-21...25), spindle orient, Z to tool-change position (zone confirms PRS-55/PRS-66), tool unclamp (SOL-10 via RLY-3/4, confirms PRS-8/PRS-9), arm swap (ATC motor FWD/REV), clamp, and retract.

Rules:

- Never reach into the changer while hydraulics are pressurized.
- If an ATC cycle stalls mid-sequence, do **not** jog axes — note the step, consult Vol 5 §8 (ATC recovery).

- Load tools in holders per the Tooling Manual (M00T002) with correct pull studs; a wrong stud can release under load.
- Keep the magazine tool table (`tool.tbl`) in sync with physical pockets — a mismatch is the most common ATC crash cause.

Chapter 7 — Running G-code Programs

1. Load the program (File → Open). The machine runs standard **EIA/ISO G-code** — Mazatrol conversational programs do **not** run directly; reprogram or post-process from CAM.
2. Set work offsets (G54...) by touching off each axis.
3. Set tool length offsets in the tool table for every tool used.
4. Verify with the preview plot; check extents against fixtures.
5. First run: **Single Block ON, rapid override down, feed override down**, hand on Feed Hold.
6. Production runs only after a proven first article.

Chapter 8 — Coolant, Air & Auxiliary Functions

Function	Command	Hardware
Flood coolant	M8 / M9	7i84U OUT11 → pump relay
Way lube	Auto/manual	7i84U OUT12; low-level switch IN25
Air blast	Custom M-code	7i97T SSR OUT0 → RLY-5 → SOL-62
Touch-sensor blast	Custom M-code	7i97T SSR OUT1 → RLY-6 → SOL-35
Tap coolant blast	Custom M-code	7i97T SSR OUT2 → RLY-7 → SOL-61
Alarm light/horn	Automatic on fault	7i84U OUT15

Coolant level (IN26) and lube level (IN25) warnings must be corrected promptly — the ways depend on the lube system (Vol 5 §3).

Chapter 9 — Feed Hold, Single Block & Cycle Control

- **Feed Hold** (panel button IN29 or GUI pause): decelerates axes, spindle keeps running. Resume with Cycle Start.
- **Single Block** (IN30): one block per Cycle Start press — use for first articles and recovery.
- **Cycle Start** (IN28 or GUI): starts/resumes the loaded program.
- **E-stop**: hardware chain — drives and hydraulics drop immediately. Use it whenever in doubt; recover via Ch. 2 power-up steps 6–7 and re-home.

Chapter 10 — Optional WHB04B Pendant

A WHB04B-style wireless USB pendant connects through LinuxCNC HAL (`pendant_whb04b.hal`), not through Mesa I/O. It provides axis select, step size, MPG wheel jogging, and cycle buttons. The pendant is enabled only after base-machine safety is proven. Pendant E-stop (if fitted) is a *soft stop* — the wall/panel E-stops remain the hardware chain.

Chapter 11 — Shutdown Sequence

1. Finish or abort the program; M5 spindle stop; M9 coolant off.
2. Move Z up and park X/Y at a convenient location.

3. Press E-stop (drops drive enables), exit LinuxCNC, shut down the PC.
4. Turn off the main disconnect if the machine will be idle.
5. Log any alarms or anomalies in the machine log for maintenance.

Chapter 12 — Operator Alarms & First Response

Symptom	Likely source	First response
Machine On refused	E-stop chain open (TB5.10)	Walk the chain: buttons, doors, overtravel finals
Joint following error	Resolver feedback / drive fault	Do not re-enable repeatedly; see Vol 5 §6
Drive fault input (X/Y/Z)	TRA drive alarm (IN10–IN12)	Read drive LED code; Vol 5 §4
Spindle won't start	Gear not confirmed, FR-SX fault (IN14), enable chain	Check gear confirm, VFD display; Vol 5 §7
ATC stalled	Interlock not satisfied	Note the step; Vol 5 §8 — never jog out blindly
Limit tripped	Overtravel	Jog off under limit-override per GUI procedure, then re-home

End of Volume 2. Continue to Volume 3 — Mesa Conversion & Installation.